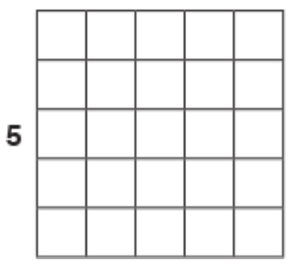
Lessons

Exponents and Roots

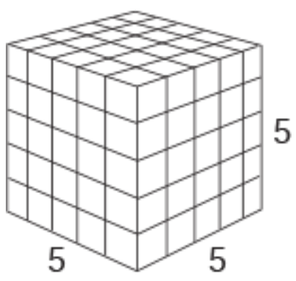
Repeated multiplication of the same number can be shown using a **power**. The expression 5^2 is read "5 to the second power." The number 5 is the **base** and 2 is the **exponent**. To solve a power, multiply the base by itself the number of times shown by the exponent.



The 5 columns and 5 rows form a square.

$5^2 = 5 \times 5 = 25$ $5^3 = 5 \times 5 \times 5 = 125$

A number raised to the second power is often called a **square.** A number raised to the third power is called a **cube.** Can you see why?



The depth is also 5. Now we have a cube.

Example 1 Hide

5^3

$5 \times 5 \times 5 = 25 \times 5 = \mathbf{125}$

Example 2 Hide

3^4

$3 \times 3 \times 3 \times 3 = \mathbf{81}$

Example 3 Hide

2^5

$2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$

The opposite of raising a number to the second power is finding the **square root**. The symbol for this operation is called a **radical sign**($\sqrt{}$). This is the symbol for a cube root: $\sqrt[3]{27}$. The cube root of 27 is 3: $3 \times 3 \times 3 = 27$.

Example 4 Hide

$\sqrt{81} = ?$

Step 1

Ask yourself, "What number multiplied by itself equals 81?"

The square root of 81 is 9.

$\sqrt{81} = \mathbf{9}$

Check: To check a square root, multiply the answer by itself. $9 \times 9 = 81$, so $\sqrt{81} = \mathbf{9}$

Many numbers do not have a whole number as a square root. To find the square root of a large number, you can use a calculator. Press the square root key and then the number. Press enter to display the square root.

If you use a calculator to find the square root of a number that is not a perfect square, the answer will have a decimal part. If you don't have a calculator, you can estimate the square root of a number by comparing it to the square roots you do know.

Example 5 Hide

Find the square root of 110.

You know that 10^2 is 100 and 11^2 is 121. Therefore, you can conclude that $\sqrt{110}$ is between the values of 10 and 11.

Key Terms

- **base**
- **exponent**
- **perfect square**
- **power**
- **scientific notation**
- **square root**

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Category

Lesson 1: Exponents and Roots

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